

Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name: Jyotirmoy Dutta				Subject: SEPM lab
TE- 1	Batch: T13	Sem: 6	Roll No: 2301051	Date: 19-01-2026

Experiment No: 02

Aim: To understand Version Control System / Source Code Management, install git and create a GitHub account

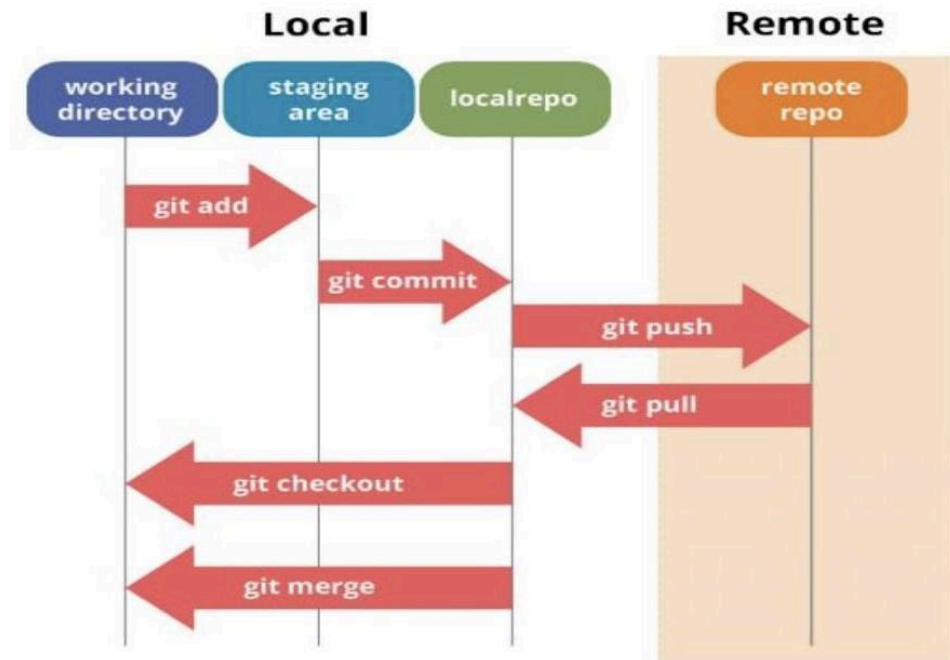
Theory:

- Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.
- Some of the basic operations in Git are:
 1. Initialize
 2. Add
 3. Commit
 4. Pull
 5. Push
- Some advanced Git operations are:
 1. Branching
 2. Merging
 3. Rebasing

The following diagram depict the all supported operations in GIT

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- **Installation of GIT**

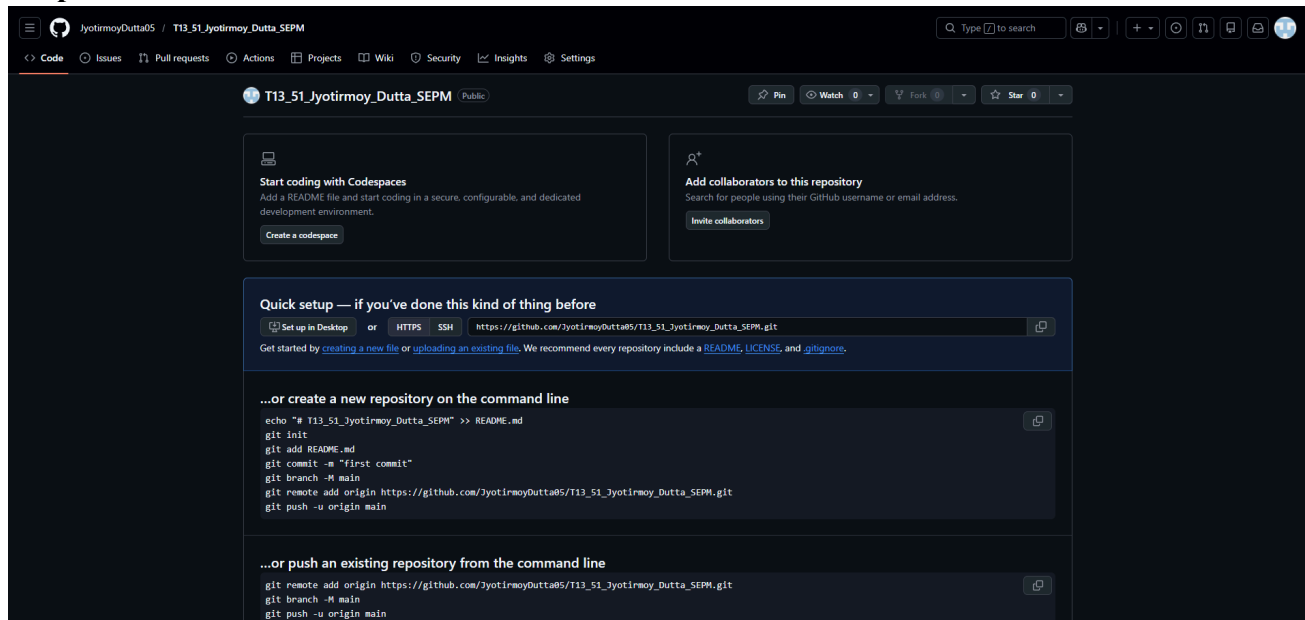
- 1) In windows, download GIT from <https://git-scm.com/> and perform the straightforward installation.
- 2) In Ubuntu, install GIT using `$sudo apt install git`, Confirm the version after installation using command `$git -version`

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Output:



```
Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git init
Reinitialized existing Git repository in C:/Users/Lab203/Documents/T13_51_Jyotirmoy_Dutta_SEPM/.git/

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git add .

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git commit -m "Initial commit"
On branch master
nothing to commit, working tree clean

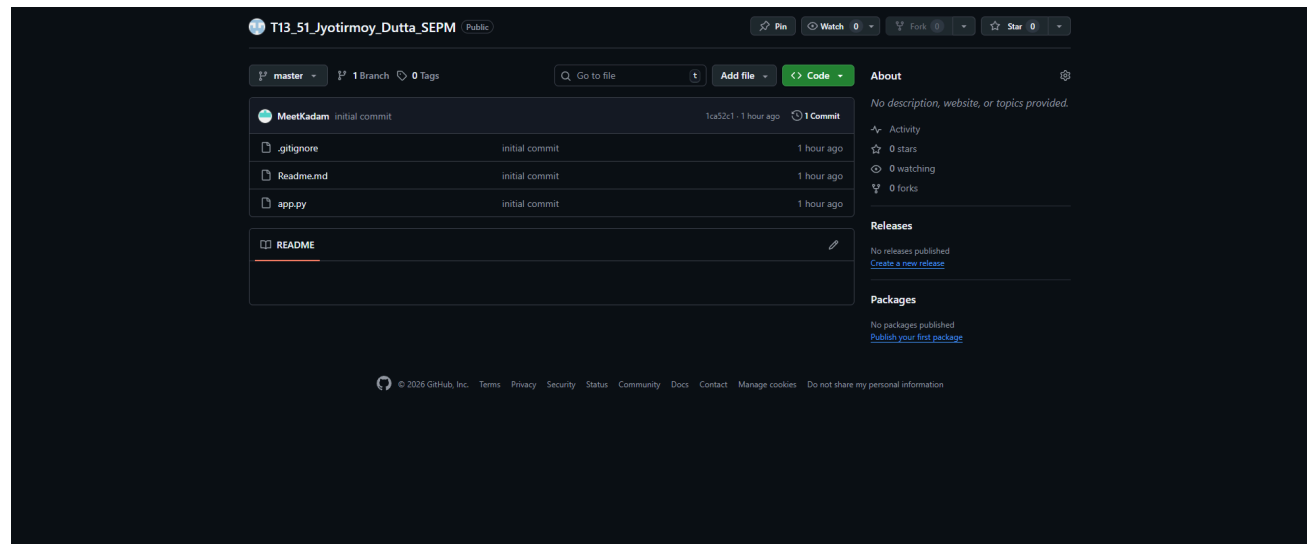
Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git branch -M master

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git remote add origin https://github.com/JyotirmoyDutta05/T13_51_Jyotirmoy_Dutta_SEPM.git
error: remote origin already exists.

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git push -u origin master
info: please complete authentication in your browser...
[]
```

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```
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 231 bytes | 231.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/JyotirmoyDutta05/T13_51_Jyotirmoy_Dutta_SEPM.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git config --global user.name "Jyotirmoy Dutta"

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git config --global --list
user.name=Jyotirmoy Dutta
user.email=meetkadam1812@gmail.com

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git config --global user.email "jyotirmoydutta05@gmail.com"

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git config --global --list
user.name=Jyotirmoy Dutta
user.email=jyotirmoydutta05@gmail.com

Lab203@L8-026 MINGW64 ~/Documents/T13_51_Jyotirmoy_Dutta_SEPM (master)
$ git push --amend --reset-author
error: unknown option `amend'
usage: git push [<options>] [<repository> [<refspec>...]]

    -v, --[no-]verbose      be more verbose
    -q, --[no-]quiet        be more quiet
    --[no-]repo <repository>
                           repository
    --[no-]all              push all branches
    --[no-]branches         alias of --all
    --[no-]mirror           mirror all refs
    -d, --[no-]delete       delete refs
    --[no-]tags             push tags (can't be used with --all or --branches or --mirror)
    -n, --[no-]dry-run      dry run
    --[no-]porcelain        machine-readable output
    -f, --[no-]force        force updates
    --[no-]force-with-lease[=<refname>:<expect>]
```

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Conclusion: Successfully performed Version Control System / Source Code Management, install git and create a GitHub account

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